Drone generation	Period/stage	Key challenges faced	Technological/strategic response
		human operators • Poor obstacle detection and sensor accuracy • Limited automation in mission tasks	and high-resolution cameras • Introduction of onboard computing and flight stabilization algorithms
4th generation (intelligent networked UAVs)	2010s–2020s	• Cybersecurity vulnerabilities • Battery and power limitations • Integration with civilian airspace	• Application of AI for mission optimization • Swarm coordination algorithms • Development of hybrid power and secure communication systems
5th generation (AI-driven and	2020s–Present	• Energy efficiency and sustainability issues	• Deployment of deep learning and self-learning algorithms